Farkle, Machine Learning, and Evidence

A result can be real without being a winner

COMSC 2043 ▪ Discrete Structures and Critical Thinking

Week 16, Tuesday, December 1, 2026

The question behind the number

PRINCIPLE: EVIDENCE BEFORE INTERPRETATION

Today's question

When a Farkle strategy appears better, what are we actually justified in believing?

- A simulation result is evidence produced by a fixed model and a finite sample.
- A machine-learning result is not a magic permission slip to say "best."
- Our job is to make the claim no larger than the evidence.

CHECKABLE: A result can be useful and still be too small or too narrow to support a broad claim.

Tuesday's path: predict, inspect, limit

1. Fix the rules and comparison.
2. Define **better** and the decision rule before the numbers.
3. Predict what the first sample will show.
4. Inspect raw wins, starts, games, and denominators.
5. Name one way a first result could mislead.
6. Write one limited conclusion and an exit trace.

Course method

Sources → Rules / Assumptions → Work → Check → One-Sentence Summary

Words we need before we read results

Farkle	A dice game in which a player banks turn points or risks them by rolling again; a Farkle loses the unbanked turn points.
strategy	A fixed rule for choosing ROLL or BANK from the state the strategy can see.
simulation	Repeatedly running a model to observe outcomes under stated rules.
sample	The particular games observed in one run.
seed	A label that makes one pseudo-random sequence repeatable.
win rate	Wins divided by the games counted for that strategy. The games are its denominator.
decision rule	A predeclared test that says what evidence is enough for a decision.

What is fixed before any game is counted?

PRINCIPLE: RULES / ASSUMPTIONS

- Target score: **4000**.
- Strategy A: `learner:500`.
- Strategy B: `bank_at_425`.
- Same classroom rules and same opponent setup.
- Each seed sample has **5** starts for A and **5** starts for B.

Fair comparison

+ one strategy label
+ the other strategy label
+ a named seed
+ a stated number of games
+ the same rules

CHECKABLE: If a rule or starting balance changes, we are no longer reading the same comparison.

Why define “better” first?

Non-code analogy

Before a race, agree whether the winner is the first to the finish line, the fastest average lap, or the least fuel used.

- “Better” is an ordering: it tells us how to rank two choices.
- A higher number is not automatically the right number.
- Choosing the metric after seeing the result lets the result choose the rule.

PREDICT (30 s): If A wins more games but takes longer to prepare, which one is better?

Our decision rule comes before the evidence

Worked rule for this deck

Call A better than B only if A has more wins than B in **every tested seed row**, with the starts balanced. If any row reverses or ties, refuse to declare a winner under this rule.

- Primary metric: observed wins in a fixed number of games.
- Scope: these strategy definitions, these classroom rules, and these tested seeds.
- This rule does not claim to find a universal best strategy.

PROMPT FOR CLAUDE

prompts/01_claim_and_rule.md

```
Sources:
- Use the fixed rules, strategy labels, and comparison setup shown in the slide.
- Do not invent a new game, opponent, metric, or data source.

Rules / Assumptions:
- State what stays fixed and define what "better" means for this comparison.
- Write a decision rule that can return "declare A," "declare B," or "refuse."

Work:
- Write one precise claim you are testing.
- Write the rule you will use before you inspect the repeated seed bundle.
- Make one prediction and give one reason for it.

Check:
- Could another reader apply your rule without guessing what "better" means?
- Does your rule name the metric, denominator, and scope?

One-Sentence Summary:
- Finish: "Under these rules, I will call ___ better only if ___; otherwise I will ___."

Do not write a Farkle engine, simulation, or machine-learning algorithm. Make the reasoning trail visible instead.
```

Predict before the first sample

Prediction

Before seeing the initial row, predict which strategy will have more observed wins under our rule, and give one reason.

My prediction: _____

Reason: _____

PREDICT

(30 s): Say the prediction aloud. It may be wrong; invisible predictions cannot be checked.

Initial sample: keep the raw counts visible

PRINCIPLE: RAW COUNTS + DENOMINATORS

Games	Starts A	Starts B	Wins A	Wins B	Ties	Win A	Win B	A–B pp
10	5	5	3	7	0	30%	70%	-40

What the row says

The sample contains **10** games, with balanced starts. The win rate is wins divided by games, not a floating impression.

CHECKABLE: Every percentage on this slide can be rebuilt from the visible wins and games.

Read the first result narrowly

Supported by this row

Under the fixed rules and seed 17, B had more observed wins than A in this sample.

Not supported by this row

That B is generally better, best against every opponent, or proven by one sample.

PROMPT FOR CLAUDE

prompts/02_initial_sample_audit.md

```
Sources:
- Use the initial evidence row shown on the slide and the fixed course setup.

Rules / Assumptions:
- Name the seed, game count, balanced starts, strategy labels, and target/rules.
- Define the denominator for each win rate.

Work:
- Copy the raw wins, starts, games, ties, and both win rates.
- State what appears better in this one sample under your predeclared rule.
- Name one specific reason this result could mislead you.

Check:
- Rebuild each rate from its numerator and denominator.
- Separate "observed in this sample" from "generally better."

One-Sentence Summary:
- Write one sentence that reports the bounded observation without declaring a universal winner.

Do not write code or design a Farkle engine. Audit the evidence and its interpretation.
```

One reason the first result might mislead

- A sample is one finite slice of random game outcomes.
- The seed makes the slice repeatable; repeatable is not the same as universal.
- Balanced starts address one fairness concern, not every source of variation.
- A learned strategy can only use the state and training experience the model gives it.

Useful sentence

This result is evidence about this bounded experiment, not a proof about all possible games.

Tuesday's limited conclusion

PRINCIPLE: SCOPE THE CONCLUSION

Exit-trace shape

In this sample, under these fixed rules, _____, but I am not yet willing to claim _____ because _____.

- Name what was observed.
- Name the condition that makes it true.
- Name the missing check or limit.

CHECKABLE: A strong answer can refuse to declare a winner while still making a precise observation.

Thursday preview: the same comparison, more seeds

Seed	Games	Starts A	Starts B	Wins A	Wins B	Ties	Win A	Win B	A–B pp
17	10	5	5	3	7	0	30%	70%	-40
2026	10	5	5	8	2	0	80%	20%	60
6262	10	5	5	6	4	0	60%	40%	20

- Same strategies, target, game count, and balanced starts.
- Different named seed rows.
- The A–B column changes sign across the rows.

PROMPT FOR CLAUDE prompts/03_bundle_stability_check.md

```
Sources:
- Use the three deterministic seed rows shown in the Thursday-preview table.

Rules / Assumptions:
- Confirm that the strategies, target, game count, classroom rules, and start balance stay fixed.
- Reuse the decision rule you wrote before seeing the bundle.

Work:
- Record each row's raw wins, denominators, win rates, and A-minus-B difference.
- Check whether the ordering has the same sign in every row.
- Find the smallest difference, largest difference, and range.

Check:
- Does the bundle satisfy your predeclared rule?
- What can the bundle support, and what would still be an overclaim?

One-Sentence Summary:
- Defend, revise, or refuse the winner claim in one evidence-bounded sentence.

Do not write a simulator or machine-learning algorithm. Compare and interpret the supplied evidence.
```

A polished interpretation can still overclaim

Confident reader / AI claim

The learner won more games in the simulation, so it is clearly the best Farkle strategy. The result proves that machine learning beats the hand-written rule.

- Hidden denominator: which games and which starts were counted?
- Missing ordering criterion: what does “best” mean?
- Overreach: one sample becomes a universal claim.
- Missing model limit: this learner and this opponent are not all possible games.

PROMPT FOR CLAUDE prompts/04_repair_an_overclaim.md

```
Sources:
- Audit the confident reader/AI statement on the slide against the displayed rows and the fixed course setup.

Rules / Assumptions:
- Treat "best" as undefined until an ordering criterion is named.
- Treat a finite simulation as evidence about this model, not proof about every possible Farkle game.

Work:
- Mark the hidden denominator, missing ordering criterion, unsupported leap from one sample to a universal claim, and ignored model limitation.
- Rewrite the statement so it says only what the evidence earns.
- Name one additional check that would be useful without turning this into a programming project.

Check:
- Does the repaired sentence include scope, rules, denominator or raw counts, and a limitation?
- Does it avoid calling a strategy "best" without a decision rule?

One-Sentence Summary:
- Finish: "The evidence supports ___ under ___, but it cannot establish ___ because ___."

Do not write a Farkle engine, simulation, or machine-learning algorithm. Repair the claim, not the software.
```

The evidence trail we carry forward

PRINCIPLE: CHECKABLE WORK

Sources
What package, CSV, or course material did we use?

Rules / Assumptions
What stayed fixed? What is each denominator?

Work
What did we predict and observe?

Check
What could mislead us? Did the bundle support the rule?

One-Sentence Summary
What bounded claim will we defend, revise, or refuse?

CHECKABLE: Thursday reopens this trail; it does not replace it with a single average.

Exit trace

Before you leave

Write one sentence in this shape:

In this sample, under these rules, _____, but I am not yet willing to claim _____ because _____.

- Include a condition.
- Include a check or limit.
- Keep the claim no bigger than the evidence.

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